Line Legend

- Process Line
- Utility Line
- Future or Existing Line
- Jacketed Line

Instrument Line Legend

- Instrument Connection Line
- Pneumatic Signal Line
- Electrical Signal Line
- Hydraulic Signal
- Capillary Tubing
- Electromagnetic, Sonic, Optical or Nuclear Signal
- Data Link or Software
- Mechanical Link

Service Fluid Designation

AV	-	Atmospheric Vent
BA	-	Breathing Air
BD	-	Blowdown
BFW	-	Boiler Feed Water
CA	-	Compressed Air
CC	-	Contaminable Condensate
CHBR	-	Chilled Brine Return
CHBS	-	Chilled Brine Supply
CHWR	-	Chilled Water Return
CHWS	-	Chilled Water Supply
CON	-	Condensate
CWR	-	Cooling Water Return
CWS	-	Cooling Water Supply
CSA	-	Concentric Sulphuric Acid
DMW	-	Demineral Water
DR	-	Drain
DS	-	Dilution Steam
DW	-	Drinking Water
DSA	-	Dilute Sulphuric Acid
EF	-	Effluent
ER	-	Ethylene Refrigerant
ETH	-	Ethanol
FLW	-	Filter Water
FG	-	Fuel Gas
FO	-	Fuel Oil
FW	-	Fire Water
HOS	-	Hot Oil Supply
HOR	-	Hot Oil Return
H2	-	Hydrogen
HCL	-	Hydrochloride Acid
HDT	-	HSD Additive
H2S	-	Hydrogen Sulphide
HPC	-	High Pressure Condensate
HPS	-	High Pressure Steam
HSD	-	High Speed Diesel
HWS	-	Hot Water Supply
HWR	-	Hot Water Return
IA	-	Instrument Air
LDO	-	Light Diesel Oil
LNG	-	Liquified Natural Gas
LO	-	Lube Oil
LPG	-	Liquified Petroleum Gas
LPC	-	Low Pressure Condensate
LPS	-	Low Pressure Steam
MDT	-	MS Additive
ME	-	Methanol
MR	-	Mixed Refrigerant
MPC	-	Medium Pressure Condensate
MPS	-	Medium Pressure Steam
MS	-	Motor Spirit
N2	-	Nitrogen
NA	-	Caustic
NG	-	Natural Gas
NH	-	Ammonia
OWS	-	Oil Water Sewer
OX	-	Oxygen
PA	-	Plant Air
PR	-	Process Fluid
PW	-	Process Water
PV	-	Process Vapour
PUW	-	Purified Water
RO	-	Reveres Osmosis
RW	-	Raw Water
RV	-	Relief Vent
SHS	-	Super Heated Steam
SLP	-	Slope Oil
SO	-	Seal Oil
SS	-	Sanitary Sewer
STS	-	Storm Sewer
SW	-	Soft Water
SWR	-	Sea Water Return
SWD	-	Sea Water Supply
SWS	-	Storm Water Drain
THF	-	Thermic fluid
TWR	-	Tempered Water Return
TWS	-	Tempered Water Supply
TW	-	Treated Water
V	-	Vent
VC	-	Vacuum Condensate
VE	-	Vacuum Exhaust
WS	-	Service Water
WW	-	Waste Water

Valves

- Gate Valve (GV)
- Check Valve (CH)
- Stop Check Valve (SCH)
- Globe Valve (GLV)
- Butterfly Valve (BF)
- Needle Valve (NV)
- Ball Valve (BV)
- Generic Rotary 1/4 Turn (GR)
- Plug Valve (PL)
- Diaphragm Valve (DV)
- **Three Way** Valve
- Four Way Valve
- Pinch Valve (PV)
- Angle Valve
- Knife Valve (KV)

Piping Specialty Items

- Y-Type Strainer
- Cone Type Strainer
- T-Type Strainer
- Duplex Strainer
- Basket Strainer
- Temporary Strainer
- Filter
- Detonation Arrestor
- Flame Arrestor
- Steam Trap Assembly
- Ejector / Eductor
- Pulsation Dampener
- In-Line Silencer
- In-Line Silencer
- Removable Spool
- Desuperheater
- Flexible Hose
- Expansion Joint
- Damper
- Breather
- Vent Cover
- In-Line Mixer
- Diverter Valve
- Rotary Valve
- Excess Flow Valve
- Exhaust Head
- Sight Glass(SG) (Double Window)
- Sight Glass(SG) (Tubular)
- Silica Gel(SGL)

Piping Fittings

- Flange
- Welded Connection
- Cap
- Reducer (Concentric)
- Reducer (Eccentric)
- Hose Connection
- Spacer
- Blank
- Spectacle Blind (Normally Open)
- Spectacle Blind (Normally Close)
- Plug
- Blind Flange
- Load Cell

Primary Element Symbol (Flow)

- General Symbol In-Line Element XX=FE, FT.etc
- In-Line Flow Element with Integral Transmitter xxxx=Mass, Coriolis, Thermal, Int. Orifice...
- In-Line Flow Element with Separate Transmitter xxxx=Mass, Coriolis, Thermal...
- Orifice
- Magnetic Meter
- Turbine or Propeller
- Ultrasonic
- Vertex
- Pitot tube
- Averaging Pitot tube
- Flow Nozzle
- Venturi
- Wedge Meter
- Flume
- Weir
- Positive Displacement
- Orifice in Quick Change Fitting
- Target
- Rotameter (or Purge) (show with Optional Integral Valve)
- Flow Switch * = H/L
- Flow Glass
- Flow Conditioning Devices (e.g., Straightening Vanes)
- Restriction Orifice

Self - Actuated Devices

- Rupture Disc or Safety Head for Pressure Relief (Explosion Panel)
- Rupture Disc or Safety Head for Vacuum Relief (Explosion Panel)
- Pressure Relief or Safety Valve
- Vacuum Relief Valve
- Pr. & Vacuum Relief Valve or Conservation Vent
- Pilot Oper.Relief Valve with Remote Sensor (use Appropriate Relief Valve Symbol)
- Pressure Reducing Regulator (Self Contained)
- Back Pressure Regulator (Self Contained)
- Pressure Reducing Regu. (W/ External Tap)
- Back Pressure Regu. (W/ External Tap)
- Differential Pressure Reducing Regulator
- Temperature Regulator Filled System Type

Control Valve Actuator Symbols

- Manual Operator
- Diaphragm
- Pressure Balanced Diaphragm
- Handwheel - Used with any Actuator
- Cylinder, Piston
- Motor Operated
- Electro - Hydraulic
- Single Solenoid
- Single Solenoid with Remote Reset
- Single Solenoid with Manual Reset
- Typical Control Valve with Positioner and Solenoid
- Goose Neck

Instrument Location

Discrete Instruments	DCS	PLC	Discrete Hardware Interlock	
○	▢	◇	◇	Field or Locally Mounted
⊖	▢	◇		Primary Location Normally Accessible to an Operator
⊖	▢	◇		Primary Location Normally Inaccessible to an Operator (Central or Main Control Room)
⊖	▢	◇		Auxiliary Location Normally Accessible to an Operator (Secondary or Local Control Room)
⊖	▢	◇		Auxiliary Location Normally Inaccessible to an Operator (Secondary or Local Control Room)

Abbreviation (General)

AG	-	Above Ground	MOV	-	Motor Operated Valve
A/M	-	Auto/Manual	MTL	-	Material
ATM	-	Atmosphere	MW	-	Manway
BL	-	Battery Limit	NC	-	Normally Closed
BTL	-	Bottom Tangent Line	NNF	-	Normally No Flow
BYP	-	Bypass	NO	-	Normally Open
CC	-	Chemical Cleanout	NOZ	-	Nozzle
CL	-	Centerline	OD	-	Outside Diameter
CO	-	Cleanout	O/C	-	Open/close
CONN	-	Connection	O/O	-	On/Off
CSC	-	Car Seal Close	OP	-	Output
CSO	-	Car Seal Open	OSBL	-	Outside Battery Limits
CTR	-	Center	OVHD	-	Overhead
DCS	-	Distribution Control Sys.	PB	-	Push Button
DES	-	Design	PLC	-	Programmable Logic Controller
DIA	-	Diameter	PRESS	-	Pressure
DP	-	Design Pressure	PV	-	Process Variable
D/P	-	Differential Pressure	(R)	-	Relocated
DRN	-	Drain	REQD	-	Required
DT	-	Design Temperature	RTD	-	Resistance Temp. Detector
DWG	-	Drawing	ROV	-	Restriction Orifice
(E)	-	Existing	SC	-	Sample Connection
EXT	-	External	SCH	-	Schedule
EL	-	Elevation	SD	-	Shoutdown
ESD	-	Emergency Shoutdown	SC	-	Specific Gravity
FA	-	Flame Arrestor	SIS	-	Safety Instrumented System
FOF	-	Face of Flange	SO	-	Steam Out
(F)	-	Furnished	SOV	-	Solenoid Operated Valve
FC	-	Fail Closed	SP	-	Sample Point
FI	-	Fail Indeterminate	SS	-	Stainless Steel
FL	-	Fail Lock (last Position)	S/S	-	Start/Stop
FLG	-	Flange	STR	-	Strainer
FO	-	Fail Open	STD	-	Standard
FP	-	Full Port	T/C	-	Thermocouple
FV	-	Full Vacuum	TDH	-	Total Differential Head
GD	-	Gas Detector	TEMP	-	Temperature
GO	-	Gear Operated	THRD	-	Threaded
GR	-	Grade	TL	-	Tangent Line
HC	-	Hose Connection	TSO	-	Tight Shout-Off
HDR	-	Header	T/T	-	Tangent to Tangent
HH	-	Hand Hole	TYP	-	Typical
HOA	-	Hand/Off/Automatic	TW	-	Thermowell
HP	-	High Pressure	UG	-	Underground
HPT	-	High Point	VNT	-	Vent
HS	-	Hand Switch	VAC	-	Vacuum
ID	-	Inside Diameter	VB	-	Vertex Breaker
ISBL	-	Inside Battery Limit	VFD	-	Variable Frequency Drive
LC	-	Lock Close	VG	-	Vacuum Gauge
LO	-	Lock Open	WT	-	Load Cell
MAX	-	Maximum	XI	-	Trip Indicator
MIN	-	Minimum	XL	-	Indication Lamp
FB	-	Full Bore	XV	-	ON/OFF Valve
HOV	-	Hand Operated Valve	XS	-	Selector Switch
PGS	-	Pig Signaler	XSL	-	Local/Remote Switch
			ZSO/C	-	Open/close Limit Switch

Instruments Identification

Pressure

PA	-	Pressure Alarm
PCV	-	Pressure Control Valve
PE	-	Pressure Element
PG	-	Pressure Gauge
PI	-	Pressure Indicator
PIC	-	Pressure Indicating Controller
PS	-	Pressure Switch
PT	-	Pressure Transmeter
PRC	-	Pressure Recorder And Controller
DPI	-	Differential Pressure Indicator
DPS	-	Differential Pressure Switch

Temperature

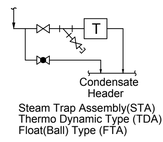
TA	-	Temperature Alarm
TCV	-	Temperature Control Valve
TE	-	Temperature Element
TG	-	Temperature Gauge
TI	-	Temperature Indicator
TIC	-	Temperature Indicating Controller
TS	-	Temperature Switch
TT	-	Temperature Transmeter
TRC	-	Temperature Recorder & Controller

Flow

FCV	-	Flow Control Valve
FE	-	Flow Element
FI	-	Flow Indicator
FIC	-	Flow Indicating Controller
FIQ	-	Flow Indicator and Totaliser
FRC	-	Flow Recorder And Controller
FS	-	Flow Switch
FT	-	Flow Transmeter
FQC	-	Flow Intergrator And Controller

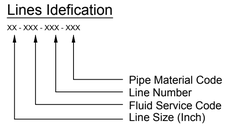
Level

LA	-	Level Alarm
LCV	-	Level Control Valve
LE	-	Level Element
LG	-	Level Gauge
LI	-	Level Indicator
LIC	-	Level Indicating Controller
LS	-	Level Switch
LT	-	Level Transmeter



Steam Trap Assembly(STA)
Thermo Dynamic Type (TDA) & Float(Ball) Type (FTA)

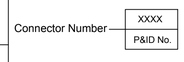
Lines Idefication



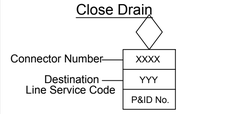
Insulation/Tracing Type

IH	-	Insulation Hot
IC	-	Insulation Cold
IP	-	Insulation for Protection
IJ	-	Steam Jacketed
IE	-	Electrical Traced
IT	-	Steam traced
CC	-	Cold Conservation
3LPE	-	3 Layer Polyethylene Layer
ICH	-	Cold & Hot Insulation

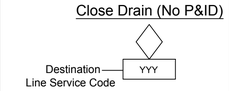
Utility Connector



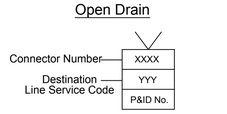
Close Drain



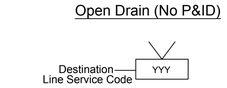
Close Drain (No P&ID)



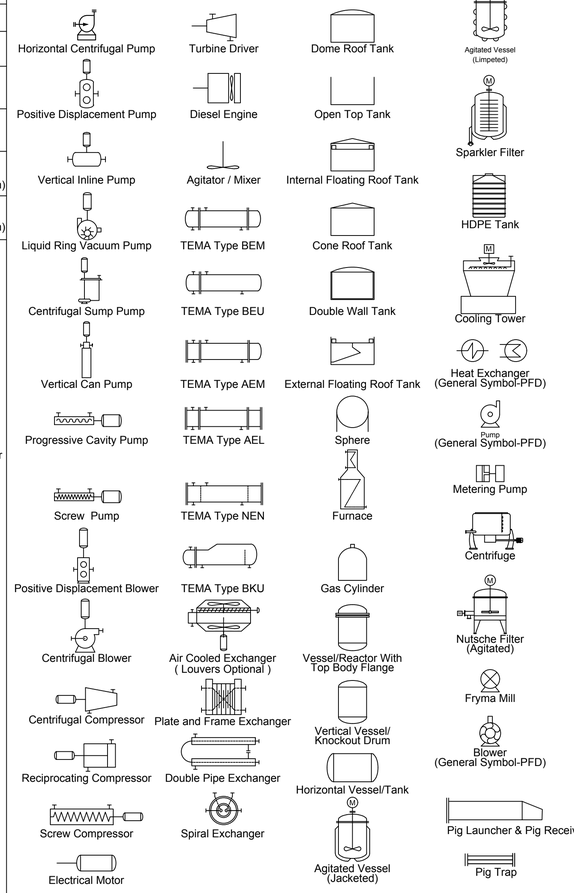
Open Drain



Open Drain (No P&ID)



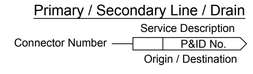
Equipment Symbol



Classification of Equipment

Class	Subject	Description
A	Mixing Equipment	Agitator, Aerator, Mechanical Mixers
B	Blowers	Centrifugal Blowers, Positive Displacement Blowers, Fans
C	Compressors	Centrifugal, Reciprocating, Screw, Vacuum
D	Mechanical Drivers	Electric and Pneumatic Motors, Diesel Engines, Steam and Gas Turbines
E	Heat Exchangers	Unfired Heat Exchangers, Condensers, Coolers, Reboilers, Vaporizers and Heating Coil, Double Pipe, Spiral, Plate & Frame, Air Coolers
F	Furnaces	Fired Heater, Furnaces, Boiler, Kilns
P	Pumps	Horizontal and Vertical Centrifugal, Positive Displacement, Vertical Canned, Screw, Gear, Sump
R	Reactors	
T	Tower / Columns	
TK	Tanks	API atmospheric and low pressure
U	Miscellaneous Equipment	Filters, Bins, Silos
V	Drums	Separators, Driers, Accumulators

Primary / Secondary Line / Drain



Miscellaneous

- XXXX — Purge Connection (XXXX = Purge Pressure and Medium)
- XX/YY — Sample Connection (XX/YY = Type/Number)
- Note Reference Symbol (XX = Note Number, Rotate Arrow as Required)
- Tie-In Symbol (XXXX = Identification Number)
- Point of Change In Line Class or Insulation Requirement
- SP-XXXX — Piping Specialty Item
- Packaged Equipment Limits
- Battery Limits (or Match Line)
- Chemical Seal / Diaphragm Pilot Light or Gauge Illuminator
- Dual Function or Instruments
- Sharing Common Housing
- Interlock

NOTES:-

Rev	Description	By	Signed	Date
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
B	ISSUED FOR APPROVAL	-	-	12.10.2021
A	ISSUED FOR REVIEW	-	-	08.09.2021

0 20 40 80 120mm
SCALE 1:2 (A1) SCALE 1:4 (A3)

0 10 20 40 60mm
SCALE 1:1 (A1) SCALE 1:2 (A3)

-

[redacted]

Drawn	Check
MS	DVR
Designed	Verified
TDR	VR

IE / PROOF CONSULTANT

Drawing Title

POL STORAGE TERMINAL AT KANECH, PUNJAB

-

PHASE 1 LEGEND SHEET

P & I DIAGRAM

Client Drawing No.	Consultant/Contractor Drawing No.	Rev
HW037-350-E-PR-DG-002	-	B